

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2015

Max. Marks: 100

Class: SE

Name of the Course: Database Management System

Course Code: UCEC404

Duration: 3 Hours

Semester: IV

Branch: COMP

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Explain the following Relational Algebraic operations with examples: (i) Cartesian product (ii) Generalized Projection (iii) Set Difference (iv) Outer Join (v) Rename	10
Q 1 (b)	Given the following relational schema Division(div#, div_name,director) Department(dept#,dept_name,location,div#) Employee(emp#,emp_name,salary,address,dept#) State the following queries in SQL i) Get the employee name,dept_name and div_name for all employees whose salary is above 20,000/- ii) List the name of all employees who work in "Marketing" division. iii) List the dept_name and employee names in that dept, for all departments whose location is Mumbai. OR Construct EER diagram for car- insurance company whose customers own one or more car each. Each car has associated with it zero or any number of recorded accidents.	03 03 04 10
Q2 (a)	Explain the steps for converting ER Data model to Relational Data Model with proper examples.	10
Q2 (b)	Consider a relation 'R' with attributes A,B,C,D,E. R={A,B,C,D,E}. The following dependencies hold; A→B BC→E ED→A. i) List all keys ii) Is R in 1NF iii) Is R in BCNF	10
Q3 (a)	Discuss the different security and authorization mechanisms in database management system.	10
Q3 (b)	Explain type of constraints with an example.	10

Q4 (a)	What is transaction? Explain ACID property of Transaction.	10
	OR Explain log based recovery technique in detail.	10
Q4 (b)	What is deadlock? What are the conditions necessary for deadlocks? Discuss the various strategies used for handling deadlock.	10
	OR What is recoverable schedule? Why is recoverability of schedules desirable? Are there any circumstances under which it would be desirable to allow non-recoverable schedules? Explain.	10
Q5 (a)	Write a short note on : i) Steps in Query Processing ii) Cost Based Query Optimization.	10
Q5 (b)	What are the levels of abstraction offered by Database management system?	10
	OR What do you mean by Database Management System? Why would you choose a database system instead of simply storing the data in operating system files?	10